# Lebesgue Inteagral

This section contains the following topics:

- Lebesgue Integral for simple functions:
  - linearity and monotonicity
  - finite additivity over measurable subsets
- Lebesgue Integral for bounded measurable functions on $E$ of finite measure
  - linearity and monotonicity
  - finite additivity over measurable subsets
  - Uniform Convergence: if $f_n \to f$ uniformly on $E$, then $\lim_{n\to\infty} \int_E f_n = \int_E f$
  - Bounded Convergence: if $f_n \to f$ a.e. on $E$, and $\{f_n\}$ are bounded, then $\lim_{n\to\infty} \int_E f_n = \int_E f$
- Lebesgue Integral for non-negative functions on a general $E$
  - linearity and monotonicity
  - finite additivity over measurable subsets
  - Fatou's Lemma: if $f_n \to f$ a.e. on $E$, then $\int_E f \leq \liminf \int_E f_n$
  - Monotone Convergence: if $f_n \to f$a.e. on $E$, and $\{f_n\}$ are increasing, then $\lim_{n\to\infty} \int_E f_n = \int_E f$
- Lebesgue Integral for general case
  - linearity and monotonicity
  - finite additivity over measurable subsets
  - Lebesgue Dominated Convergence: if $f_n \to f$ a.e. on $E$, and $|f_n| \leq g$, and $g$ is integrable over $E$, then $\lim_{n\to\infty} \int_E f_n = \int_E f$
  - General Lebesgue Dominated Convergence: if $f_n \to f$ a.e. on $E$, and $|f_n| \leq |g_n|$, and $g_n$ is integrable over $E$, and $\lim_{n\to\infty} \int_E g_n = \int_E g < \infty$, then $\lim_{n\to\infty} \int_E f_n = \int_E f$
- Countable Additivity and Continuity

## Lebesgue Integral for Simple Functions

Suppose a simple $\phi$, defined on a measurable set $E$, take values $a_1, \ldots, a_n$. Let $E_i = \phi^{-1}(a_i), i = 1, \ldots, n$. Then, define the **Lebesgue Integral** of $\phi$ over $E$ as

$$\int_E \phi = \sum_{i=1}^{n} [a_i \cdot m(E_i)]$$

**Lemma 1.10.1: Let $\{E_i\}_{i=1}^n$ be a finite disjoint collection of measurable subsets of $E$, which is of finite measure. For $1 \leq i \leq n$, let $a_i$ be a real number. Define $\phi = \sum_{i=1}^n a_i \cdot \mathcal{X}_{E_i}$. Then**

$$\int_E \phi = \sum_{i=1}^{n} a_i \cdot m(E_i)$$

*Proof:* The trick is re-arrange and group $E_i$ by the same value $a_i$.

**QED**

**Proposition 1.10.2 (Linearity): Let $\phi$ and $\psi$ be two simple functions on a common domain $E$. For any real numbers $\alpha, \beta$, we have**

$$\alpha \cdot \int_E \phi + \beta \cdot \int_E \psi = \int_E (\alpha \cdot \phi + \beta \cdot \psi)$$

*Proof:* If $\alpha = 0$, then clearly $0 \cdot \int_E \phi = \int_E 0 \cdot \phi$. Suppose $\phi$ takes values $a_1, \ldots, a_n$ on $E_1, \ldots, E_n$, respectively. Then $\alpha \cdot \phi$ takes values $\alpha \cdot a_1, \ldots, \alpha \cdot a_n$ on $E_1, \ldots, E_n$. So

$$\int_E \alpha \cdot \phi = \sum_{i=1}^{n} [\alpha \cdot a_i \cdot m(E_i)] = \alpha \sum_{i=1}^{n} [a_i \cdot m(E_i)] = \alpha \cdot \int_E \phi$$

Now it remains to show $\int_E (\phi + \psi) = \int_E \phi + \int_E \psi$.

Suppose $\phi$ takes values $a_1, \ldots, a_n$ on $E_1, \ldots, E_n$, and $\psi$ takes values $b_1, \ldots, b_m$ on $D_1, \ldots, D_m$. Now let $C_{i,j} = E_i \cap D_j$. Then the simple function $\phi + \psi$ takes values $a_i + b_j$ on $C_{i,j}$. So by Lemma 1.10.1, we have

$$(1) \quad \int_E (\phi + \psi) = \sum_{1 \le i \le n,\ 1 \le j \le m} (a_i + b_j) \cdot m(C_{i,j})$$

On the other hand, note that

$$(2) \quad \int_E \phi = \sum_{1 \le i \le n} a_i \cdot m(E_i) = \sum_{1 \le i \le n} \sum_{1 \le j \le m} a_i \cdot m(C_{i,j})$$

, and

$$(3) \quad \int_E \psi = \sum_{1 \le j \le m} b_j \cdot m(D_j) = \sum_{1 \le j \le m} \sum_{1 \le i \le n} b_j \cdot m(C_{i,j})$$

. By combining (1)-(3) we obtain the result as desired.

**QED**

**Proposition 1.10.3 (Monotonicity): $\phi$ and $\psi$ are two simple functions defined on a common domain $E$. If $\phi \le \psi$, then $\int_E \phi \le \int_E \psi$.**

*Proof:* Define $\eta = \phi - \psi$. Since $\phi \le \psi$, we have $\eta \le 0$. It is easy to see that $\eta$ is simple function. So by the linearity, we have

$$\int_E \phi - \int_E \phi = \int_E (\phi - \psi) = \int_E \eta \le 0.$$

**QED**

# Lebesgue Integral for Bounded Measurable Functions over finite measure $E$

Let $f$ be a **bounded function** defined on a measurable set $E$, **which is of finite measure**.

Define the **lower Lebesgue integral** of $f$ over $E$ as

$$\sup\{\int_E \phi \mid \phi \leq f,\ and\ \phi\ is\ simple\}$$

Define the **upper Lebesgue integral** of $f$ over $E$ as

$$\inf\{\int_E \psi \mid f \leq \psi,\ and\ \psi\ is\ simple\}$$

If the two integrals equal, then we say $f$ is **Lebesgue integrable** over $E$, and denote its integral by

$$\int_E f = \sup\{\int_E \phi \mid \phi \leq f,\ and\ \phi\ is\ simple\} = \inf\{\int_E \psi \mid f \leq \psi,\ and\ \psi\ is\ simple\}$$

**Proposition 1.10.4: Let $f$ be a bounded measurable function on $E$ of finite measure. Then $f$ is integrable over $E$.**

*Proof:* Since $f$ is measurable, by Simple Approximation Lemma, for any $n$ there are simple functions $\phi_n$ and $\psi_n$ such that $\phi_n \leq f \leq \psi_n$, and $|\phi_n - \psi_n| < \frac{1}{n}$. For convenience, let $\overline{\int_E} f$ be it supper integral, and let $\underline{\int_E} f$ be its lower integral. It is easy to see that

$$\int_E \phi_n \leq \underline{\int_E} f \leq \overline{\int_E} f \leq \int_E \psi_n,\ for\ every\ n$$

. So

$$(1)\quad \overline{\int_E} f - \underline{\int_E} f \leq \int_E \psi_n - \int_E \phi_n$$

By linearity of integrals for simple functions, (1) becomes

$$\overline{\int_E} f - \underline{\int_E} f \leq \int_E (\psi_n - \psi_n) < \frac{1}{n} \cdot m(E) \to_{n\to\infty} 0$$

**QED**

**Proposition 1.10.5 (Linearity and Monotonicity): Let $f, g$ be bounded measurable functions on finite measure $E$. For any $\alpha, \beta$, we have**

$$\alpha \cdot \int_E f + \beta \cdot \int_E g = \int_E (\alpha \cdot f + \beta \cdot g)$$

If $f \leq g$ on $E$, then

$$\int_E f \leq \int_E g$$

*Proof:* First of all, by Proposition 1.10.4, the integrability of $\alpha \cdot f$, $\beta \cdot g$, and $\alpha \cdot f + \beta \cdot g$ are guaranteed because they are measurable on $E$.

The case $\alpha = 0$ is obvious. Suppose $\alpha > 0$. Let $A_1 = \{\int_E \phi \mid \phi \leq f\}$ and $A_2 = \{\int_E \phi' \mid \phi' \leq \alpha \cdot f\}$.

We claim $\alpha \cdot \sup\{A_1\} = \sup\{A_2\}$, and thus establish $\alpha \cdot \int_E f = \int_E \alpha \cdot f$.

In fact, for any $\phi \in A_1$, there is an simple function $\phi' = \alpha \cdot \phi$ such that $\alpha \cdot \phi \leq \phi' \leq \alpha \cdot f$. Therefore, $\alpha \cdot \sup\{A_1\} \leq \sup\{A_2\}$. On the other hand, for any $\phi' \in A_2$, there is a $\phi = \frac{1}{\alpha}\phi' \leq f$ such that $\phi' \leq \alpha\phi$ . Therefore, $\sup\{A_2\} \leq \alpha \cdot \sup\{A_1\}$.

The case when $\alpha < 0$ can be proved similarly.

Now it remains to show that $\int_E f + \int_E g = \int_E (f+g)$. Let $A_f = \{\int_E \phi_f \mid \phi_f \leq f\}$ and $A_g = \{\int_E \phi_g \mid \phi_g \leq g\}$. For any $\phi_f \in A_f$ and $\phi_g \in A_g$, we have $\phi_f + \phi_g \leq f + g$. So

$$(1) \quad \int_E (f+g) \geq \int_E (\phi_f + \phi_g) = \int_E \phi_f + \int_E \phi_g$$

. Note (1) is true for every $\phi_f \in A_f, \phi_g \in A_g$, and $f, g$ are integrable. Therefore, we must have

$$(2) \quad \int_E (f+g) \geq \int_E f + \int_E g$$

. Similarly we have prove that

$$(3) \quad \int_E (f+g) \leq \int_E f + \int_E g$$

For monotonicity, if $f \leq g$ on $E$, by linearity we have

$$\int_E f - \int_E g = \int_E (f-g) \leq 0$$

**QED**

**Proposition 1.10.6 (Finite Additivity): Let $f$ be a bounded measurable function on $E$ of finite measure. Let $A \subseteq E$ be a measurable subset. Then,**

$$\int_E f = \int_A f + \int_{E-A} f$$

*Proof:* Define $f_1 = f \cdot \mathcal{X}_A$, and $f_2 = f \cdot \mathcal{X}_{E-A}$. Both $f_1, f_2$ are bounded and measurable on $E$. And $f_1 + f_2 = f$. By linearity, we have

$$\int_E f = \int_E (f_1 + f_2) = \int_E f_1 + \int_E f_2 = \int_A f + \int_{E-A} f$$

. The last equality follows from the fact that

$$\int_E f \cdot \mathcal{X}_{\mathcal{A}} = \int_A f$$

, which is easy to prove.

**QED**

**Proposition 1.10.7 (Uniform Convergence): Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of measurable functions on a common domain $E$ of finite measure. Every $f_n$ is bounded on $E$. If $f_n \to f$ uniformly on $E$, then**

$$\int_E f = \lim_{n\to\infty} \int_E f_n$$

*Proof:* Since $f_n$ converges to $f$ uniformly, $f$ is measurable. Furthermore, since every $f_n$ is bounded, $f$ is also bounded on $E$. As a result, $\int_E f$ exists. So by linearity we have

$$\int_E f - \int_E f_n = \int_E (f - f_n)$$

for every $n$.

Since $f_n$ converges to $f$ uniformly, for any $\epsilon > 0$ there is a sufficiently large $N$ such that whenever $n \geq N$, we have

$$|f(x) - f_n(x)| < \frac{\epsilon}{m(E)}$$

. Thus,

$$\int_E (f - f_n) \leq \int_E |f - f_n| < \frac{\epsilon}{m(E)} \cdot m(E) = \epsilon$$

, when $n$ is sufficiently large. As a result, the limit

$$\lim_{n\to\infty} (\int_E f - \int_E f_n) = 0$$

**QED**

If the convergence is point-wise in Proposition 1.10.7, the exchange of $\int$ and $\lim$ is not justified.

Consider the following counter-example: $E = [0, 1]$. $f = 0$ on $E$. $f_n(x) = n$ for $x \in [0, \frac{1}{n}]$ and $f_n(x) = 0$ otherwise. Then $\int_E f = 0$. But every $\int_E f_n = 1$.

**Proposition 1.10.8 (Bounded Convergence):** Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of measurable functions on a common domain $E$ of finite measure. Suppose $\{f_n\}$ are **uniformly bounded** on E, i.e. there exists $M > 0$ such that $|f_n(x)| \leq M$ for every $x \in E$ and every $n$. If $f_n \to f$ a.e. on $E$, then $\int_E f = \lim_{n\to\infty} f_n$

*Proof:* Note that $m(E) < \infty$. Therefore, we can apply Egoroff's Theorem. For any $\epsilon > 0$, there is a closed subset $F \subseteq E$ such that $m(E - F) < \frac{\epsilon}{4M}$, and $f_n \to f$ uniformly on $F$. So for sufficiently large $n$, we have $|f_n - f| < \frac{\epsilon}{2 \cdot m(F)}$ for all $x \in F$. Therefore, for sufficiently large $n$,

$$\begin{aligned}
\int_E |f_n - f| &= \int_F |f_n - f| + \int_{E-F} |f_n - f| \\
&\leq \frac{\epsilon}{2 \cdot m(F)} \cdot m(F) + \int_{E-F} |f_n| + \int_{E-F} |f| \\
&\leq \frac{\epsilon}{2} + M \cdot \frac{\epsilon}{4M} + M \cdot \frac{\epsilon}{4M} \\
&= \epsilon
\end{aligned}$$

**QED**

# Lebesgue Integral for Non-negative functions on a general measurable set $E$

Let $h$ be a measurable function on $E$. We say that $h$ has **finite support**, if there exists a measurable subset $E_0 \subseteq E$ such that $m(E_0) < \infty$ and $h(x) = 0$ for $x \in E - E_0$.

If a function $h$ is bounded and has finite support, we define its Lebesgue integral as

$$\int_E h = \int_{E_0} h$$

With the notion of finite support, we can now define Lebesgue integral for a non-negative function on a general measurable set $E$ as

$$\int_E f = \sup\{\int_E h \mid h \; bounded, \; measurable \; and \; of \; finite \; support, \; 0 \leq h \leq f\}$$

**Proposition 1.10.9 (Chebyshev's Inequality): Let $f$ be a non-negative function on $E$. Then for any $\lambda$, we have**

$$m(\{x \in E \mid f(x) \geq \lambda\}) \leq \frac{1}{\lambda} \cdot \int_E f$$

*Proof:* Let $E_\lambda = \{x \in E \mid f(x) \geq \lambda\}$.

Suppose $m(E_\lambda) = \infty$. Take $E_{\lambda,n} = E_\lambda \cap [-n, n]$ where $n$ is a natural number. Then take $h_n = \lambda \cdot \mathcal{X}_{E_{\lambda,n}}$. Then $h_n$ is bounded, measurable and of finite support. So $\int_E h_n = \lambda \cdot m(E_{\lambda,n})$. By the continuity of measure, we have

$$\infty = \lim_{n \to \infty} \lambda \cdot m(E_{\lambda,n}) = \lambda \cdot m(E_\lambda) = \lim_{n \to \infty} \int_E h_n$$

Now note that $\int_E h_n$ is increasing and unbounded. Since $\int_E f \geq \sup\{\int_E h_n\}$, thus $\int_E f = \infty$. The conclusion holds.

Next, suppose $m(E_\lambda) < \infty$. Take $h = \lambda \cdot \mathcal{X}_{E_\lambda}$. Then $h \leq f$, and is bounded, measurable and of finite support. So

$$\lambda \cdot m(E_\lambda) = \int_E h \leq \int_E f$$

**QED**

**Proposition 1.10.10: Let $f$ be a non-negative measurable function on $E$. Then, $\int_E f = 0$, if and only if, $f = 0$ a.e. on $E$.**

*Proof:* If $\int_E f = 0$, by Chebyshev's inequality, we have

$$m(\{x \in E \mid f(x) \geq \frac{1}{n}\}) = 0$$

So taking the intersection and invoking the continuity property of measure, we have $m(\{x \in E \mid f(x) > 0\}) = 0$. That is, $f(x) = 0$ a.e. on $E$.

Conversely, if $f = 0$ a.e. on $E$, then consider any simple function $\phi$ and any function $h$ that is bounded, measurable and of finite support such that $0 \leq \phi \leq h \leq f$. It is easy to see that $\phi = 0$ a.e. on $E$. Therefore, $\int_E \phi = 0$ for any such simple functions. As a result, $\int_E h = 0$. Since $h$ is arbitrary, we imply that $\int_E f = 0$. **QED**

**Proposition 1.10.11 (Linearity and Monotonicity): $f, g$ are two non-negative measurable functions on $E$. Then for any real numbers $\alpha, \beta$, we have**

$$\alpha \cdot \int_E f + \beta \cdot \int_E g = \int_E (\alpha \cdot f + \beta \cdot g)$$

If $f \leq g$ on $E$, then

$$\int_E f \leq \int_E g$$

*Proof:* Simple and omitted.

**QED**

**Proposition 1.10.12 (Finite Additivity): Let $f$ be a non-negative measurable function on $E$. Let $A \subseteq E$ be a measurable subset. Then,**

$$\int_E f = \int_A f + \int_{E-A} f$$

*Proof:* Simple and omitted.

**Proposition 1.10.13 (Fatou's Lemma): Let $\{f_n\}$ be a sequence of non-negative measurable functions on $E$. If $f_n \to f$ a.e. on $E$, then**

$$\int_E f \leq \liminf\{\int_E f_n\}$$

*Proof:* It is easy to see that $f$ is a non-negative measurable function on $E$. To prove the conclusion, we would like to show that for any non-negative function $h$ that is bounded, measurable, of finite support and satisfies $h \leq f$, we have $\int_E h \leq \liminf\{\int_E f_n\}$.

To this end, consider such a function $h$. Let $E_0$ be its finite support. Construct a sequence of functions $h_n$ as: $h_n(x) = \min\{h(x), f_n(x)\}$. Observe that $h_n \to h$ p.w. on $E_0$. Because for any $x \in E_0$, since $f_n(x) \to f(x)$, for sufficiently large $n$: if $h(x) \leq f_n(x)$ then $h_n(x) = h(x)$; if $f_n(x) < h(x)$, then $f_n(x)$ will approach $h(x)$, for otherwise, since $h(x) \leq f(x)$, infinitely many $|f_n(x) - h(x)| > \epsilon$ would fail the point-wise convergence.

Since $h_n \to h$ on $E$, and $h_n \leq h$ are uniformly bounded, by Bounded Convergence we have

$$\int_E h = \int_{E_0} h = \lim_{n\to\infty} \int_{E_0} h_n = \lim_{n\to\infty} \int_E h_n$$

Note that $h_n \leq f_n$ on $E$ as well. So

$$\liminf\{\int_E h_n\} \leq \liminf\{\int_E f_n\}$$

**QED**

**Proposition 1.10.14 (Monotone Convergence):** Let $\{f_n\}$ be a sequence of non-negative measurable functions on $E$. If $f_n$ are increasing and $f_n \to f$ a.e. on $E$, then

$$\int_E f = \lim_{n\to\infty} \int_E f_n$$

*Proof:* By Fatou's lemma, we have $\lim_E f \leq \liminf \int_E f_n$. For the other direction, since $f_n$ are increasing and converges p.w. to $f$, we have $f_n \leq f$. So by the monotonicity we have

$$\int_E f_n \leq \int_E f$$

, for every $n$. So

$$\limsup \int_E f_n \leq \int_E f$$

**QED**

We say that a non-negative function $f$ is integrable over $E$ if

$$\int_E f < \infty$$

**Proposition 1.10.15: If $f$ is integrable over $E$, then $f$ is finite a.e. on $E$.**

*Proof:* By Chebyshev's inequality, we have

$$m(\{x \in E \mid f(x) = \infty\}) \leq m(\{x \in E \mid f(x) > n\}) \leq \frac{1}{n} \cdot \int_E f$$

for every $n$. Since $\int_E f < \infty$, we have $m(\{x \in E \mid f(x) = \infty\}) = 0$.

**QED**

**Proposition 1.10.16 (Beppo Levi's Lemma): $\{f_n\}$ is an increasing sequence of non-negative measurable functions on $E$. The sequence $\int_E f_n$ is bounded. Then $f_n \to f$ p.w. on $E$, and $f$ is finite a.e. on $E$, and**

$$\lim_{n\to\infty} \int_E f_n = \int_E f < \infty$$

*Proof:* Let $f(x) = \lim_{n\to\infty} f_n(x), x \in E$. Since $f_n$ are increasing, $f(x)$ exists (either a real value or $\infty$). Since $f_n$ are increasing, by Monotone Convergence Theorem, we have $\lim_{n\to\infty} \int_E f_n = \int_E f$. Since $\int_E f_n$ are bounded, so is $\int_E f$. Finally, by Proposition 1.10.15, $f$ is finite a.e. on $E$.

**QED**

# Lebesgue Integral for General Cases

Let $f$ be a measurable function on $E$. Then $f^+, f^-, |f|$ are non-negative functions. It is not difficult to see that

$|f|$ **is integrable over $E$, if and only if, $f^+, f^-$ are both integrable over $E$.**

So we define the Lebesgue integral for $f$ over $E$ as

$$\int_E f = \int_E f^+ - \int_E f^-$$

It is not difficult to prove the linearity and monotonicity.

**Proposition 1.10.17 (Linearity and Monotonicity): $f, g$ are two measurable functions integrable over $E$. Then for any real number $\alpha, \beta$, we have**

$$\alpha \cdot \int_E f + \beta \cdot \int_E g = \int_E (\alpha \cdot f + \beta \cdot g)$$

If $f \leq g$ on $E$, then

$$\int_E f \leq \int_E g$$

*Proof:* Not difficult.

**QED**

**Proposition 1.10.18 (Finite Additivity): $f$ is integrable over $E$. Let $E_0$ be a measurable subset of $E$. Then,**

$$\int_E f = \int_{E_0} f + \int_{E-E_0} f$$

*Proof:* Not difficult.

**QED**

If $f$ is "dominated" by another integrable function, then it is integrable as well.

**Proposition 1.10.18: $f$ is a measurable function on $E$. And $g$ is integrable over $E$. If $|f| \leq g$ on E, then $f$ is integrable over $E$.**

*Proof: * Note that both $|f|, g$ are non-negative functions. If $\int_E g < \infty$, then $\int_E |f| \leq \int_E g < \infty$.

**QED**

We have two convergence theorems for general Lebesgue integrals.

**Proposition 1.10.19 (Lebesgue Dominated Convergence Theorem): $\{f_n\}$ is a sequence of measurable functions on $E$. $g$ is a measurable function that is integrable over $E$. Suppose $f_n$ is dominated by $g$ for every $n$. That is, $|f_n| \leq g$ on $E$ for every $n$. Then, if $f_n \to f$ a.e. on $E$, then $f$ is integrable over $E$ and**

$$\lim_{n\to\infty} \int_E f_n = \int_E f$$

*Proof:* Since $|f_n| \leq g$ and $f_n \to f$ a.e. on $E$, we know $|f| \leq g$ a.e. on $E$. Since $g$ is integrable over $E$, so are $f$ and $f_n$ for every $n$. As a result, $f_n, f, g$ are finite a.e. on $E$. So we may assume $f_n, f, g$ are finite on $E$, and $f_n \to f$ p.w. on $E$.

Now consider the non-negative functions $g - f$ and $g - f_n$. Then $g - f_n \to g - f$ p.w. on $E$. By Fatou's Lemma, we have

$$\int_E g - \int_E f = \int_E (g - f) \leq \liminf \int_E (g - f_n) = \int_E g - \limsup \int_E f_n$$

Thus, we have

$$\limsup \int_E f_n \leq \int_E f$$

. Now consider the functions $g + f_n$ and $g + f$. Note that since $|f_n| \leq g$ and $|f| \leq g$, we know $g + f_n$ and $g + f$ are non-negative functions. Repeat the Fatou's Lemma, we have

$$\int_E f \leq \liminf \int_E f_n$$

**QED**

**Proposition 1.19.20 (General Lebesgue Dominated Convergence Theorem): $\{f_n\}$ is a sequence of measurable functions on $E$. $\{g_n\}$ is a sequence of measurable functions each of which is integrable over $E$. Suppose $f_n$ is dominated by $g_n$ for every $n$. That is, $|f_n| \leq g_n$ on $E$ for every $n$. In addition, $g_n \to g$ a.e. on $E$ and $\lim \int_E g_n = \int_E g < \infty$. Then, if $f_n \to f$ a.e. on $E$, then $f$ is integrable over $E$ and**

$$\int_E f = \lim \int_E f_n$$

*Proof:* First, since $\lim \int_E g_n < \infty$, we have $\int_E g_n < \infty$ for every $n$. Since $|f_n| < g_n$, $f_n$ is integrable over $E$ for every $n$. So, $f_n, g_n$ are finite a.e. on $E$. The functions $g_n - f_n$ and $g_n + f_n$ can be well-defined.

Now, since $|f_n| \leq g_n$, and $f_n \to f$ and $g_n \to g$, it is not difficult to see that $|f| \leq g$. As a result, $f$ is integrable over $E$. And $g - f, g + f$ can be well-defined.

Similar to the proof in Proposition 1.19.19, we apply Fatou's Lemma. Note that here we can't simply use $g - f_n$ and $g - f$, because $|f_n| \leq g$ is not true.

First, since $g_n - f_n$ and $g - f$ are all non-negative, and $g_n - f_n \to g - f$, we have

$$\int_E g - \int_E f = \int_E (g - f) \leq \liminf \int_E (g_n - f_n)$$

Therefore,

$$(1) \quad \int_E f \geq \limsup \int_E [f_n + (g - g_n)]$$

Note that $\lim \int_E (g_n - g) = 0$. So

$$(2) \quad \limsup \int_E [f_n + (g - g_n)] = \limsup \int_E f_n + \limsup \int_E (g - g_n) = \limsup \int_E f_n$$

Combining (1) and (2), we have

$$\int_E f \geq \limsup \int_E f_n$$

. Similarly, we can prove $\int_E f \leq \liminf \int_E f_n$.

**QED**

# Countable Additivity and Continuity of Lebesgue Measure

The finite additivity property can be generalized to countable additivity.

**Proposition 1.10.21: $f$ is integrable over $E$. $\{E_n\}$ is a countable collection of disjoint subsets of $E$, for which $\bigcup_{n=1}^{\infty} E_n = E$. Then**

$$\int_E f = \sum_{n=1}^{\infty} \int_{E_n} f$$

*Proof:* Define $f_n = f \cdot C_{X_n}$, where $X_n = \bigcup_{k=1}^n E_k$. Note that $|f_n| \leq |f|$, and $|f|$ is integrable over $E$. Therefore, $f_n$ is integrable for every $n$, and

$$(1) \quad \int_E f = \lim_{n\to\infty} \int_E f_n$$

However, by finite additivity, we have

$$(2) \quad \int_E f_n = \int_{\bigcup_{k=1}^n E_k} f = \sum_{k=1}^n \int_{E_k} f$$

Combining (1) and (2), we obtain the result as desired.

**QED**

**Proposition 1.10.22: Let $f$ be integrable over $E$.**

(i) $\{E_n\}$ is an increasing sequence of measurable subsets of $E$. Then

$$\int_{\bigcup_{n=1}^\infty E_n} f = \lim_{n\to\infty} \int_{E_n} f$$

(ii) $\{F_n\}$ is a decreasing sequence of measurable subsets of $E$. Then

$$\int_{\bigcap_{n=1}^\infty F_n} f = \lim_{n\to\infty} \int_{F_n} f$$

*Proof:* Note that $f$ is integrable over any $E_n, F_n$ and their unions and intersections.

For upward continuity, let $A_n = E_n - E_{n-1}, n \geq 2$ and $A_1 = E_1$. Then $\{A_n\}$ are disjoint. Note that $\bigcup_{n=1}^\infty A_n = \bigcup_{n=1}^\infty E_n$. So, by countable additivity, we have

$$\int_{\bigcup_{n=1}^\infty E_n} f = \int_{\bigcup_{n=1}^\infty A_n} f = \sum_{n=1}^\infty \int_{A_n} f = \lim_{n\to\infty} \sum_{k=1}^n \int_{A_k} f = \lim_{n\to\infty} \int_{E_n} f$$

For downward continuity, let $B_n = F_1 - F_n$. Then $B_n$ is increasing, and $\bigcap_{n=1}^\infty F_n = F_1 - \bigcup_{n=1}^\infty B_n$. So, by the result of part (i), we have

$$(3) \quad \int_{F_1 - \bigcap_{n=1}^\infty F_n} f = \int_{\bigcup_{n=1}^\infty B_n} f = \lim_{n\to\infty} \int_{B_n} f = \lim_{n\to\infty} \int_{F_1 - F_n} f$$

Therefore, by (3) we have

$$\int_{F_1} f - \int_{\bigcap_{n=1}^\infty F_n} f = \int_{F_1} f - \lim_{n\to\infty} \int_{F_n} f$$

. Finally, since $\int_{F_1} f < \infty$, we obtain the result as desired.

**QED**

# Exercise

**Q1.** Let $\{f_n\}$ be a sequence of measurable functions integrable over $E$. $f_n \to f$ a.e. on $E$ and $f$ is integrable over $E$. Show that $\int_E |f_n - f| \to 0$ if and only if $\lim_{n\to\infty} \int_E |f_n| = \int_E |f|$.

*Solution:* From left to right: Let $g_n = |f_n - f|$, and $h_n = |f_n| - |f|$. Then $g_n \to 0$ a.e. on $E$ and $h_n \to 0$ a.e. on $E$. Note that $|h_n| \leq g_n$ on $E$. Clearly, every $g_n$ is integrable over $E$. So if $\int_E g_n \to 0$, then by General Lebesgue Convergence Theorem, we have $\lim_{n\to\infty} \int_E h_n = 0$.

Conversely, note that $0 \leq |f_n - f| - (|f_n| - |f|) \leq 2|f|$. Since $|f|$ is integrable, by Lebesgue Dominated Convergence Theorem, $|f_n - f| - (|f_n| - |f|)$ is integrable, and

$$\lim_{n\to\infty} \int_E [|f_n - f| - (|f_n| - |f|)] = 0$$

Since $\lim_{n\to\infty} \int_E (|f_n| - |f|) = 0$, we have the result as desired.

**QED**

**Q2.** $f(x, y)$ is a real-valued function that is defined on $\{(x, y) \mid 0 \leq x \leq 1,\ 0 \leq y \leq 1\}$. $f(x, y)$ is a measurable function of $x$ for each fixed value $y \in [0, 1]$. Suppose $\lim_{y\to 0} f(x, y) = f(x)$ for each $x \in [0, 1]$ and that for all $y$, we have $|f(x, y)| \leq g(x)$, where $g(x)$ is integrable over $[0, 1]$. Show that

$$\lim_{y\to 0} \int_0^1 f(x, y)dx = \int_0^1 f(x)dx$$

If $f(x, y)$ is continuous on $y$ for every $x$, show that

$$h(y) = \int_0^1 f(x, y)dx$$

is also continuous on $y \in [0, 1]$

*Solution:* Let $f_n(x) = f(x, \frac{1}{n})$. Then $f_n \to f$ p.w. on $E$, and $|f_n(x)| \leq g(x)$. Since $g(x)$ is integrable, by Lebesgue Dominated Convergence, we have

$$\lim_{n\to\infty} \int_0^1 f_n(x)dx = \int_0^1 f(x)dx$$

. In addition, if $f(x, y)$ is continuous at $y$, then $f(x, y_n) \to f(x, y)$ for any sequence $y_n \to y$. Again, $|f(x, y_n| \leq g(x)$ for any $x, y_n$. So by Lebesgue Dominated Convergence, we have

$$\lim_{n\to\infty} h(y_n) = \lim_{n\to\infty} \int_0^1 f(x, y_n)dx = \int_0^1 f(x, y)dx = h(y)$$

. So $h(y)$ is continuous at $y$.

**QED**

**Q3.** $f(x,y)$ is a real-valued function that is defined on $\{(x,y) \mid 0 \le x \le 1,\ 0 \le y \le 1\}$. $f(x,y)$ is a measurable function of $x$ for each fixed value $y \in [0,1]$. For each $(x,y)$, the partial derivative $\frac{\partial f}{\partial y}$ exists. Suppose there is a function $g(x)$ that is integrable over $[0,1]$, such that

$$|\frac{\partial f}{\partial y}(x,y)| \le g(x),\ \ for\ all\ x, y$$

Show that

$$\frac{d}{dy}\Big[\int_0^1 f(x,y)dx\Big] = \int_0^1 \frac{\partial f}{\partial y}(x,y)dx\ \ for\ all\ y \in [0,1]$$

*Solution:* Consider any $y \in [0,1]$. Define $h_n(x,y) = n \cdot [f(x, y+\frac{1}{n}) - f(x,y)]$. Since the partial derivative exsits, $\lim_{n\to\infty} h_n(x,y)$ exists, and

$$\lim_{n\to\infty} h_n(x,y) = \frac{\partial f}{\partial y}(x,y)$$

Note that we would like to prove

$$(1)\ \ \lim_{n\to\infty}\int_0^1 h_n(x,y)dx = \int_0^1 \frac{\partial f}{\partial y}(x,y)dx$$

To prove (1), we claim that $h_n(x,y)$ is integrable over $[0,1]$ for every $n$ and apply Lebesgue Dominated Convergence Theorem. To this end, note that since $\frac{\partial f}{\partial y}(x,y)$ exists for all $y \in [0,1]$, $f(x,y)$is continuous at all $y$ . So by Mean Value function, for every $n$, there is some $\eta \in [y, y+\frac{1}{n}]$ such that $|h_n(x,y)| = |\frac{\partial f}{\partial y}(x,\eta)| \le g(x)$. Therefore, $h_n(x,y)$ is integrable.

**QED**